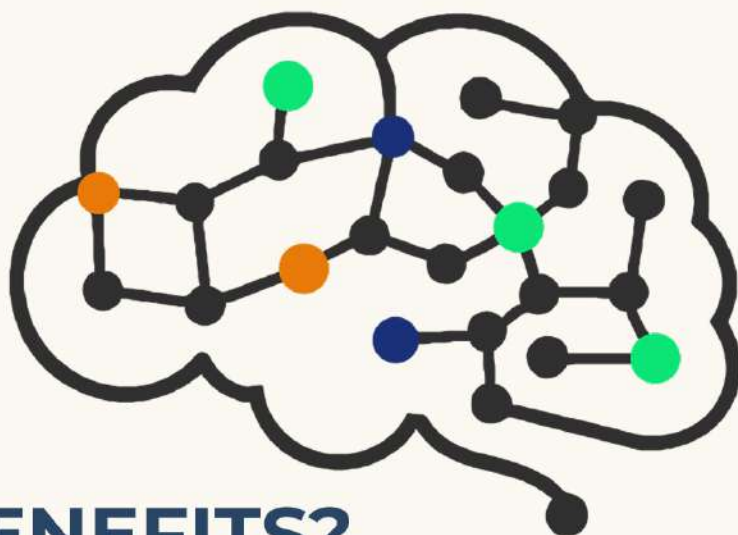


FEDERATED LEARNING

WHAT IS IT?

Collaborative training of AI models across devices, keeping data local so that models learn collectively without exposing the users' private data.



WHAT ARE THE BENEFITS?



Privacy-Preserving



Legal Compliant



Scalable



Personalised

1



MODEL DISTRIBUTION

The server packages the latest global ("mother") model and stores it in Azure Blob Storage. Edge devices poll the storage and download the model.

2



LOCAL TRAINING

Each edge device loads the model and trains it locally on its personal dataset, leveraging on-device compute.

3



UPDATE UPLOAD

Upon training completion, the edge device serializes its model updates and uploads them back to Azure Blob Storage.

4



AGGREGATION ORCHESTRATION

A Temporal workflow monitors Azure Blob Storage, collects model updates from devices, and securely aggregates them into a new global model.

5



MODEL UPDATE & LOOP

The aggregated global ("child") model is stored in Blob Storage and the workflow starts the next round by notifying the edge devices, continuing the federated learning loop.